State Machines – Part 2
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Review

- We have discussed some terminology.
- We also used them to analyse one circuit.
- We will now look at more examples of analysis of circuits involving specific types of flip-flops.

Example 2

- We will now analyse one more circuit involving D flip-flops. We will assume here that the input equation is given.
- Analysis involves drawing the state table and state diagram.

Start with the input equation ..

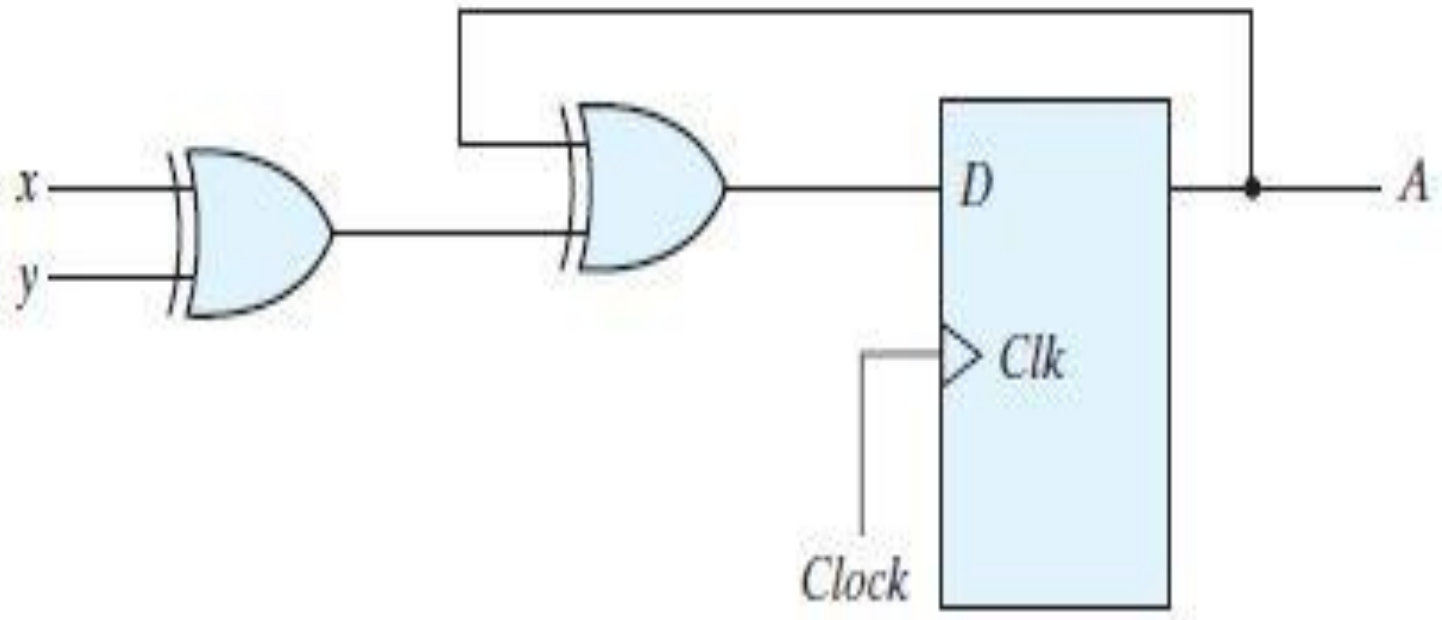
- The circuit we want to analyze is described by this input equation:

$$D_A = A \oplus x \oplus y$$

where D_A symbol implies a flip-flop with output A. The x and y variables are inputs to the circuit. No output equations are given so we assume output is that of the flip-flop.

Logic diagram from the input equation

$D = x \oplus y \oplus A$



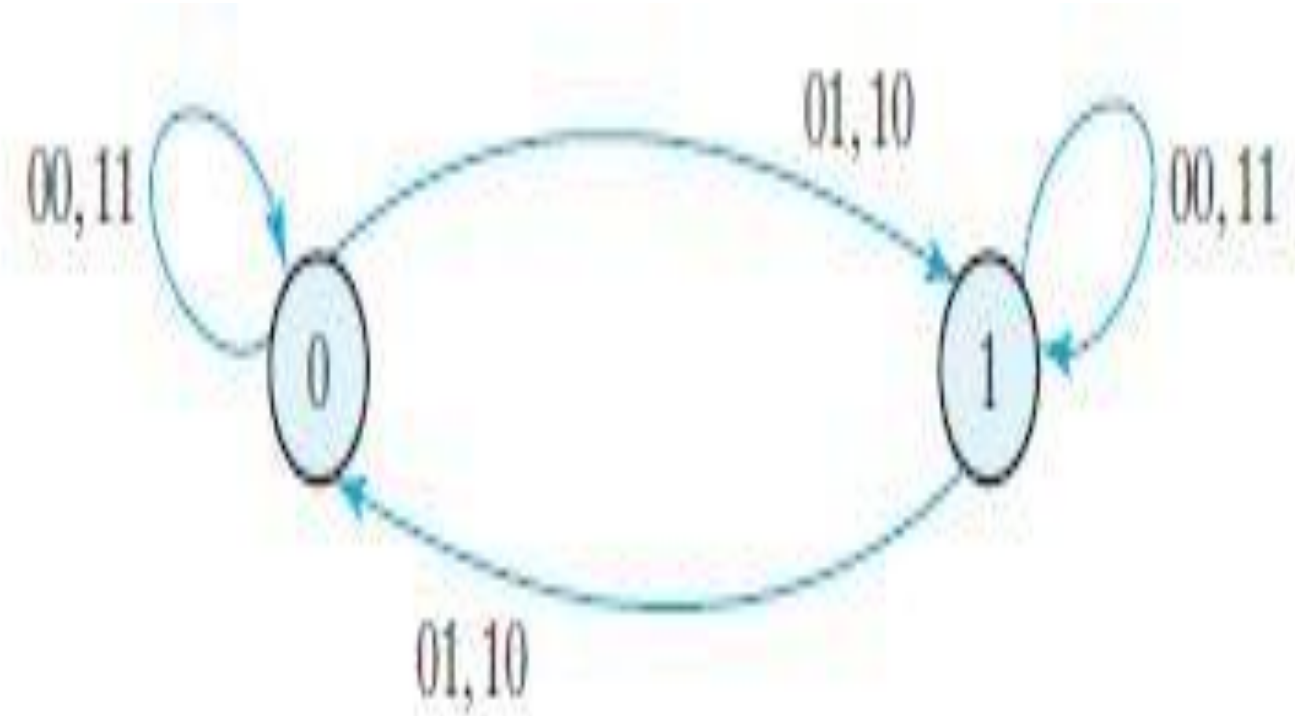
State table

Present state	Inputs		Next state
A	x	y	A
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Explanation for the state table

- A state table, in general, has four sections: present state, inputs, next state and outputs.
- In this case, there are no "explicit" outputs and hence just three sections.
- The binary numbers under A x y columns are listed from 000 to 111.
- The next state values are obtained from the state equation: A (t+1) = A xor x xor y.

State diagram



Explanation for the state diagram

- We have not put a slash (/) on the directed lines since it is not needed (no output is specified here unlike the previous example)
- Since there are two inputs (x and y), we have pairs such as 00, 11.
- The next state can be “reached” by more than one set of values for x and y (see table).

Example 3

- We will now analyze a circuit that has a pair of JK flip-flops.
- In the case of a circuit with just a D flip-flop (seen earlier), the state equation is obtained directly from the input equation.
- When we have other flip-flops (like JK or T), the procedure is slightly more elaborate (we will need to refer to the characteristic table or equation for the flip-flop for example). The details are as follows.

Procedure to get next-state values

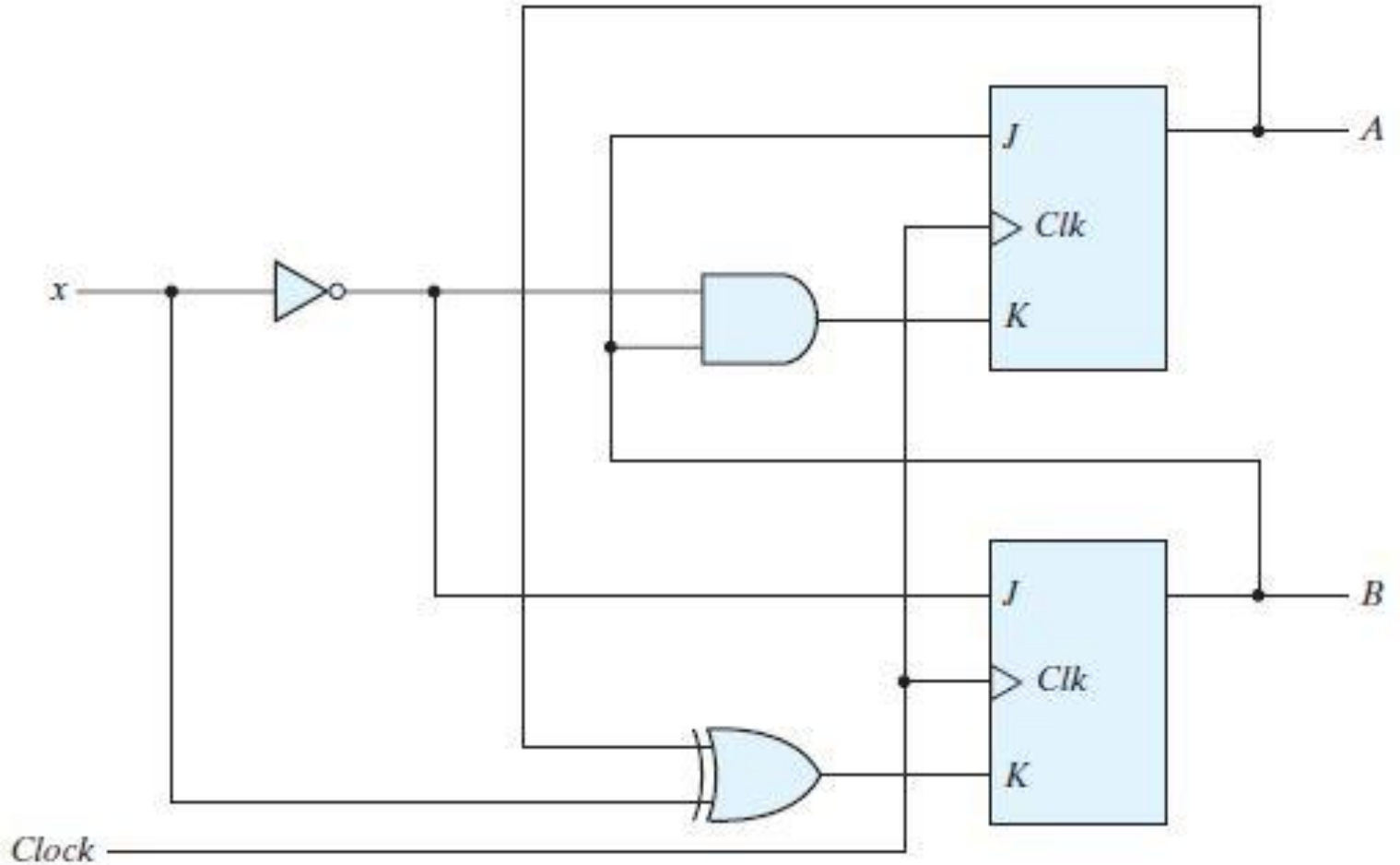
- To get the next state values for the state table (for a circuit that uses JK or T flip-flops), we do the following:

  (a) Determine the flip-flop input equations in terms of the present state and input variables.

  (b) List the binary values of each input equation.

  (c) Use the corresponding flip-flop characteristic table to determine the next state values (for the state table)

The circuit for example 3



The input equations

$$J_A = B \quad K_A = Bx'$$

$$J_B = x' \quad K_B = A'x + Ax' = A \oplus x$$

Explanation

- The flip-flop input equations can be written directly by observing the circuit given.
- There are 2 inputs per flip-flop and so altogether we have 4 equations.

State Table

Present State		Input	Next State		Flip-Flop Inputs			
A	B	x	A	B	J_A	K_A	J_B	K_B
0	0	0	0	1	0	0	1	0
0	0	1	0	0	0	0	0	1
0	1	0	1	1	1	1	1	0
0	1	1	1	0	1	0	0	1
1	0	0	1	1	0	0	1	1
1	0	1	1	0	0	0	0	0
1	1	0	0	0	1	1	1	1
1	1	1	1	1	1	0	0	0

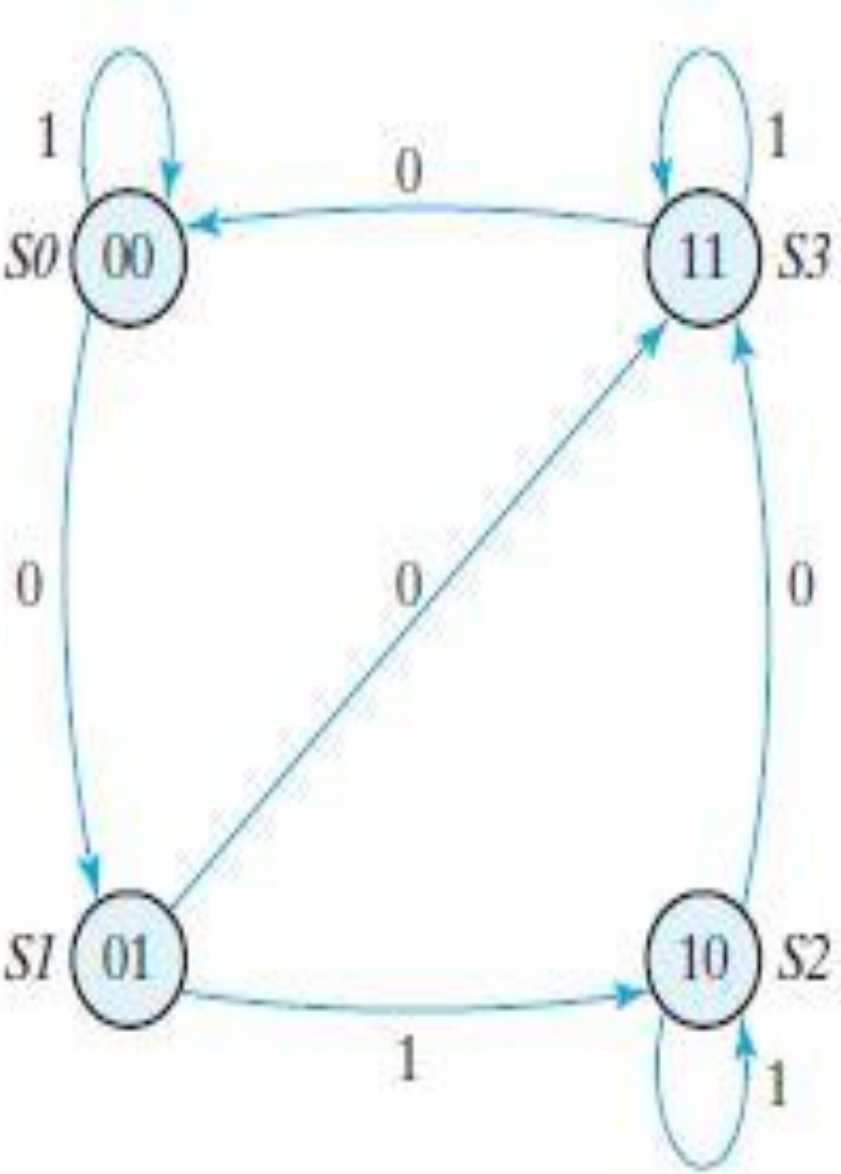
Explanation

- Truly, the state table does not include the columns labelled as “flip-flop inputs” but these columns help to get the entries in the next state column.
- We use the input equations to get the entries in the columns labelled “flip-flop inputs”.
- Then we evaluate the next state of each flip-flop using the characteristic table of the flip-flop along with the entries in the flip-flop input column. We will illustrate this next.

Illustration

- Consider row 1 of the state table. First, we need to identify the values of J_A , K_A, J_B and K_B from the present state and input values (using the input equation).
- Then use the characteristic table for JK flip-flop: we know that when J = 1 and K =0, the next state is 1. This matches the entry for next state for B (since $J_B = 1$ and $K_B = 0$)
- Now consider the entry for next state under “A” in row 1: we know that when $J_A = K_A = 0$, there is no change of state. Hence, next state entry under A is 0. Similarly, we can interpret the remaining rows.

State Diagram



Explanation

- The state diagram is straightforward to draw once we have the state table.
- Note that the directed lines out of the circles are labelled only with a single binary number (which corresponds to the value of input x).
- Two flip-flops hence we have 4 states in the diagram.

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- S. Brown and Z. Vranesic, Fundamentals of digital logic with VHDL design, McGraw Hill, 2017.